An explanation is not simply "interpretable" or "not interpretable." Explanation quality has multiple dimensions that should be evaluated separately. The framework defines **12 desirable properties**, grouped into **Content, Presentation, and User-related properties**

- **Interpretability** is the degree to which a person can understand a model or decision. An intrinsically interpretable model exposes meaningful internal variables or rules; a post-hoc explainer approximates or probes an already trained model.
- **Transparency** describes inspectability of data, transformations, model versions, rules, and provenance. It is necessary for auditability but does not itself establish faithfulness.
- **Faithfulness** asks whether the explanation tracks the computational factors that changed the model decision.
- **Fidelity** often means agreement between an explanatory surrogate and the black box. Because the literature uses fidelity and faithfulness inconsistently, this report defines every experimental metric operationally.
- **Plausibility/coherence** asks whether an explanation looks sensible to a person. A plausible explanation can be unfaithful; a faithful one can be clinically meaningless.
- **Clinical evidence** can justify why a relationship is medically credible without proving that the learned model used it. Conversely, a faithful attribution can reveal a model dependency that is clinically wrong.
- **Causal explanation** requires causal assumptions or interventions. Attribution and ordinary counterfactual model queries are not causal treatment effects.
- **Integrated Gradients:** Integrates model gradients along a linear path from a baseline input to the actual input

Explainability AI Techniques

Factual / Feature Attribution-Based Techniques
These explain *why* a model made a specific prediction by scoring the contribution of each input feature ("X caused Y").
SHAP, LIME, Integrated Gradients, Grad CAM, DeepLIFT, Permutation Feature Importance
Counterfactual & Contrastive Techniques These explain what minimum changes to the input would flip the prediction ("If X were X', then Y would become Y'").
DiCE, Growing Spheres, Contrastive Explanations Method (CEM), Actionable Recourse Optimization
Proposition-Based & Rule Extraction Techniques These translate complex model logic into logical propositions or IF-THEN rules.
Anchors, Decision Tree Surrogates(Global Rules), RuleFit, Knowledge Graph Embedding/Fuzzy Logic
Example-Based / Instance-Based Techniques These explain predictions by referencing similar training data instances rather than abstract features.
Prototypes and Criticisms, Influence Functions, Knearest Neighbors
Intrinsic / Model-Based Interpretable Architectures Instead of post-hoc explanation tools, these models are designed to be self-explaining by structure.
Generalized Additive Models, Attention Mechanisms, Concept Bottleneck Models

Concept-based explanations,
causal explanations,
response/dependence explanations,
representation/internal-state explanations,
mechanistic interpretability,
training-data attribution,
natural-language rationalization,
structural/graph explanations,
interaction explanations,
and sequential/policy explanations

XAI family	Fundamental question	Representative techniques
Feature Attribution	Which inputs contributed to Y?	SHAP, IG, DeepLIFT, PFI
Saliency / Visualization	Where did the model focus?	Grad-CAM, Score-CAM, SmoothGrad
Feature Effect / Response	How does changing X affect Y?	PDP, ICE, ALE
Feature Interaction	Which variables jointly influence Y?	SHAP Interaction, H-statistic, 2D ALE
Counterfactual / Recourse	What minimum change would change Y?	DiCE, Growing Spheres, CEM
Causal / Interventional	What actually causes Y?	SCMs, causal attribution, Shapley
Rule / Symbolic / Logical	Which understandable rules produce Y?	Anchors, RuleFit, Rule Lists, symbolic explanations
Surrogate / Approximation	Can a simple model approximate the black box?	LIME, surrogate trees
Example / Prototype	Which known examples explain this result?	kNN, prototypes, criticisms
Training-Data Attribution	Which training examples caused this behavior?	Influence Functions, TracIn, Data Shapley
Concept-Based	Which human concepts explain Y?	TCAV, ACE, ConceptSHAP, CBMs
Representation-Based	What do hidden representations encode?	probing, activation maximization, network dissection
Propagation / Decomposition	How is output relevance distributed backward?	LRP, Deep Taylor
Mechanistic / Circuit	What internal computation produced Y?	activation patching, circuits, SAEs
Graph / Structural	Which nodes/edges/subgraphs matter?	GNNExplainer, PGExplainer, SubgraphX
Knowledge / Semantic Path	What domain knowledge supports Y?	KG paths, ontologies, neuro-symbolic reasoning
Natural-Language Rationalization	Can the reason be expressed for humans?	extractive/abstractive rationales, grounded NLEs
Uncertainty-Aware	How sure is the model and why?	calibration, Bayesian uncertainty, conformal methods
Temporal / Sequential / Policy	Why this action/state at this point in time?	trajectory attribution, reward decomposition
Intrinsic Interpretable Architecture	Can the model itself be understandable?	GAM, EBM, trees, CBMs, rule lists

Proposition-Based & Rule Extraction : Symbolic / Logical / Rule / Proof-Based Explainability
Example Based : Similarity, Prototype, Criticism, Influence, Contrast

Dimension	Possible values
Scope	Local ↔ Global
Timing	Ante-hoc/Intrinsic ↔ Post-hoc
Model dependence	Model-specific ↔ Model-agnostic
Explanation target	Prediction ↔ Model ↔ Representation ↔ Dataset
Output form	Numerical ↔ Rules ↔ Examples ↔ Graph ↔ Visualization ↔ Text
Fidelity	Faithful ↔ Plausible
Audience	Developer ↔ Domain expert ↔ End user ↔ Regulator
Data modality	Tabular ↔ Image ↔ Text ↔ Graph ↔ Time series ↔ Multimodal

Group	Co-12 property
Content	**1. Correctness**
	2. Completeness
	3. Consistency
	4. Continuity
	5. Contrastivity
	6. Covariate Complexity
Presentation	**7. Compactness**
	8. Composition
	9. Confidence
User	**10. Context**
	11. Coherence
	12. Controllability